

1 The human eye responds to different light intensities.

(a) The table shows the mean pupil diameter in different light intensities.

Light intensity	Pupil diameter / mm
bright light	3.35
normal room light	3.86
darkness	6.41

Calculate the percentage increase in the mean diameter of the pupil when a person moves from bright light into darkness.

(1)

Answer %

(b) During the fight or flight response the pupils of a person increase in diameter.

(i) Which is the name of the hormone involved in the fight or flight response?

(1)

- ☐ **A** adrenaline
- ☐ **B** antidiuretic hormone
- ☐ **C** thromboplastin
- ☐ **D** uracil

(ii) Describe how this change in diameter is brought about by the muscles of the iris.

(3)

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(Total for Question 1 = 5 marks)